Albert School

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Graph Analytics - Mr. Gianluca Quercini's Teaching

Project - Graph Analytics of Collective Fish Schooling Behaviour

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Source code and experimental results :
GitHub -- Projet_Graph_Analytics

Abstract

This study investigates the structural organisation of a fish school through the construction and analysis of spatial proximity graphs. Using empirical observations of individual positions and velocities, the objective is to examine how graph-theoretical properties can describe collective cohesion, fragmentation and local organisation.

The analysis focuses on connectivity, graph distances, centrality measures and community structure. These complementary approaches provide a mathematical framework for describing the relationships between individual neighbourhoods and the overall configuration of the school.

Two complementary graph representations are considered : an undirected spatial proximity graph and a directed visibility graph. Their structural properties are compared across observation times and neighbourhood radii.

Particular attention is given to the interpretation of the resulting network measures and to the limitations introduced by the graph construction process.

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1 Introduction

A team of fish, much like a flock of birds, provides a representative example of collective behaviour in which the movements of individual agents give rise to coherent patterns at the group level. The observation of such systems reveals different forms of spatial organisation, ranging from highly coordinated motion to local fragmentation, dispersion and subsequent reaggregation.

The understanding of these collective dynamics requires a representation capable of describing both the characteristics of individual fish and the interactions that may occur between neighbouring individuals. Graph theory provides an appropriate mathematical framework for this purpose. Each fish can be represented by a vertex, while edges connect individuals satisfying a predefined spatial proximity criterion. The resulting network makes it possible to examine the local structure of the school and to investigate how individual neighbourhoods contribute to its overall organisation.

Several hypotheses may be considered to explain the configurations observed within a fish school. Collective cohesion may result from a sufficiently dense network of local interactions ; conversely, the disappearance of certain connections may produce isolated individuals or distinct connected components. Moreover, the structural position of individual fish may vary considerably, suggesting differences in their potential contribution to the connectivity and organisation of the group.

The present project therefore aims to investigate the extent to which the topological properties of spatial proximity graphs can account for the cohesion, fragmentation and structural organisation of a fish school. The analysis relies on empirical observations of individual positions and velocities, from which graphs are constructed and examined using established methods of network analysis.

Particular attention is given to connectivity, graph distances, centrality measures and community structure. These properties provide complementary descriptions of the network : connectivity characterises the extent to which individuals belong to a common structure ; distances describe the separation between vertices in terms of successive connections ; centrality measures identify structurally prominent individuals ; and community detection investigates the existence of relatively cohesive subgroups.

The relevance of these measures will ultimately be assessed through their confrontation with the observed spatial configurations of the school. It is important, however, to distinguish the mathematical properties of a proximity network from the underlying biological interactions. The presence of an edge indicates spatial closeness according to the selected construction rule ; it does not, by itself, establish communication, behavioural influence or a causal relationship between two fish.

The objective is consequently not only to calculate network statistics, but also to determine what these statistics reveal about the organisation of the observed system, while identifying the limitations introduced by the graph representation itself.

2 Dataset Presentation

2.1 Description and Source of the Dataset

The present study is based on an empirical dataset describing the movements of a school of fish. The data are obtained from the *Fish Schooling Data Subset*, made available through Oregon State University.

The dataset contains observations associated with a school of 300 fish, recorded at a frequency of 30 frames per second over 5,000 frames. It provides a temporal description of individual movements through two-dimensional positions and velocity components.

The principal variables considered in this analysis are the following :

— **px** : horizontal position of an individual fish ;
— **py** : vertical position of an individual fish ;
— **vx** : horizontal component of its velocity ;
— **vy** : vertical component of its velocity ;
— **onfish** : field used as an individual tracking identifier in the project implementation.

These variables make it possible to reconstruct the spatial configuration of the school at successive observation times and to investigate the evolution of its local neighbourhood structure.

The data are stored in JSON format. Their temporal organisation allows the analysis of individual frames as well as comparisons between distinct observation periods.

2.2 Representation of the Fish School as a Graph

The empirical observations are not initially provided as a network. A graph representation must therefore be constructed from the recorded positions of the individuals.

Let $G_t = (V_t, E_t)$ denote the graph associated with an observation frame t, where V_t represents the set of fish included in the analysis and E_t the set of connections established between neighbouring individuals.

Each vertex $i \in V_t$ is associated with a position vector

$$X_i(t) = \begin{pmatrix} x_i(t) \\ y_i(t) \end{pmatrix} \in \mathbb{R}^2, \tag{1}$$

and a velocity vector

$$V_i(t) = \begin{pmatrix} v_{x,i}(t) \\ v_{y,i}(t) \end{pmatrix}. \tag{2}$$

The spatial separation between two individuals i and j is defined by the Euclidean distance

$$d_{ij}(t) = \sqrt{(x_i(t) - x_j(t))^2 + (y_i(t) - y_j(t))^2}. \tag{3}$$

A connection is introduced whenever this distance satisfies a predefined proximity criterion :

$$(i,j) \in E_t \quad \Longleftrightarrow \quad i \neq j \quad \text{and} \quad d_{ij}(t) < R, \tag{4}$$

where $R > 0$ denotes the neighbourhood radius.

This construction produces an undirected, simple graph, provided that the proximity criterion is applied symmetrically and that each pair of fish is connected by at most one edge.

The number of vertices is denoted by

$$N_t = |V_t|, \tag{5}$$

while the number of edges is given by

$$M_t = |E_t|. \tag{6}$$

Although the original dataset describes a school of 300 fish, the effective number of vertices in a particular frame must be determined from the valid observations retained during graph construction.

The resulting network provides a representation of spatial proximity rather than a direct measurement of social or behavioural interactions. Its structure consequently depends on the selected radius, the distribution of fish positions and the completeness of the available observations.

2.3 Interpretation of the Graph Representation

The construction of a proximity graph makes it possible to move from the individual description of fish trajectories to the analysis of collective organisation.

A fish situated within a densely occupied region may be connected to numerous neighbours, whereas an individual located at the periphery of the school may have relatively few connections. Similarly, spatially separated groups may form distinct connected components, reflecting a fragmentation of the network under the selected proximity criterion.

The choice of the neighbourhood radius is therefore of particular importance. A small radius may produce a sparse and fragmented graph, while a larger radius may connect previously separated groups and increase the overall density of the network.

The subsequent analysis examines these structural properties using graph-theoretical measures, with particular attention to the interpretation of their variations and their dependence on the graph construction parameters.

3 Description of the Analysis

3.1 Analytical Framework and Software

The analysis of the fish school relies on the construction of graphs representing the relationships between individuals at a given observation time.

Two complementary models have been implemented. The first is based exclusively on the spatial proximity of the fish, while the second introduces a directional visibility constraint derived from their velocity vectors.

The objective is to examine the extent to which these representations make it possible to describe the connectivity, cohesion and internal organisation of the school.

The numerical analysis has been developed in Python, using the NetworkX library for graph construction and the calculation of network measures. NumPy and SciPy are used for numerical operations and Euclidean distance calculations, while Matplotlib provides the graphical representations of the results.

The experimental data are read from the JSON dataset using the ijson library, which allows individual frames to be extracted without loading the entire dataset into memory.

The principal methods are described below.

3.2 Construction of the Spatial Proximity Graph

The first model considers an undirected graph in which each vertex represents an individual fish and each edge corresponds to a spatial neighbourhood relationship.

For two distinct individuals i and j, the adjacency matrix is defined by

$$A_{ij}(t) = \begin{cases} 1, & \text{if } d_{ij}(t) < R, \\ 0, & \text{otherwise,} \end{cases} \qquad i \neq j, \tag{7}$$

with

$$A_{ii}(t) = 0. \tag{8}$$

The parameter R denotes the neighbourhood radius.

Since the Euclidean distance is symmetric, the resulting adjacency matrix satisfies

$$A_{ij}(t) = A_{ji}(t). \tag{9}$$

The graph is therefore undirected. It is also simple, since self-loops and multiple edges are excluded.

Each edge retains the Euclidean distance between the corresponding individuals as an attribute. However, the structural measures presented in this study are calculated without using these distances as weights.

All detected fish are retained as vertices, including those having no neighbours within the selected radius.

The reference value is set to

$$R = 100, \tag{10}$$

expressed in the native coordinate units of the dataset.

This value is used as a descriptive reference rather than as a biologically validated interaction distance.

Additional analyses are performed for

$$R \in \{60, 80, 100, 120, 140\}, \tag{11}$$

in order to examine the sensitivity of the network properties to the selected spatial threshold.

3.3 Construction of the Directed Visibility Graph

The second model introduces an additional condition associated with the direction of movement of each fish.

Unlike the spatial graph, the visibility graph is directed. An arc from individual i to individual j exists when the two fish satisfy the proximity condition and when j lies outside the rear blind sector of i.

The direction of movement is approximated by

$$\theta_i(t) = \operatorname{atan2}\left(v_{y,i}(t), v_{x,i}(t)\right). \tag{12}$$

The direction from individual i towards individual j is given by

$$\phi_{ij}(t) = \operatorname{atan2}\left(y_j(t) - y_i(t), x_j(t) - x_i(t)\right). \tag{13}$$

The angular difference is calculated using

$$\Delta\theta_{ij}(t) = \operatorname{atan2}\left(\sin(\phi_{ij} - \theta_i), \cos(\phi_{ij} - \theta_i)\right). \tag{14}$$

A directed connection is introduced if

$$d_{ij}(t) < R \tag{15}$$

and

$$|\Delta\theta_{ij}(t)| \leq \pi - \frac{\alpha}{2}, \tag{16}$$

where α denotes the total angular width of the rear blind sector.

In the present implementation,

$$\alpha = 25^\circ. \tag{17}$$

The nominal field of visibility therefore covers 335°.

The resulting graph is directed because the visibility condition is not necessarily symmetric :

$$A_{ij}(t) \neq A_{ji}(t) \tag{18}$$

for some pairs of individuals.

A fish may consequently be able to perceive another individual without the reciprocal condition being satisfied.

This representation remains a geometrical approximation. The velocity vector is used as a proxy for orientation ; the model does not directly measure the visual perception or attention of the animals.

3.4 Connectivity and Connected Components

The connectivity of a graph provides a first indication of its global organisation.

For the undirected spatial graph, two vertices are connected if a path exists between them.

A connected component is a maximal subset of vertices within which every pair of individuals can be joined by a sequence of edges.

The number of connected components is denoted by

$$C(G). \tag{19}$$

The relative size of the largest connected component is calculated as

$$S_{\mathrm{LCC}} = \frac{|V_{\mathrm{LCC}}|}{|V|}. \tag{20}$$

This quantity measures the proportion of observed individuals belonging to the largest connected structure.

For the directed visibility graph, two distinct definitions are considered :

— **Weak connectivity** : connectivity obtained when the direction of the arcs is ignored ;

— **Strong connectivity** : connectivity for which every vertex can reach every other vertex through directed paths.

These definitions are not interchangeable and must be distinguished when comparing the two graph models.

3.5 Graph Density and Degree Distribution

The degree of a vertex i in the undirected graph corresponds to the number of neighbouring fish connected to that individual :

$$k_i = \sum_{j=1}^{N} A_{ij}. \tag{21}$$

The average degree is defined by

$$\langle k \rangle = \frac{2M}{N}, \tag{22}$$

where N is the number of vertices and M the number of edges.

The density of an undirected simple graph is

$$D(G) = \frac{2M}{N(N-1)}. \tag{23}$$

For a directed graph without self-loops, the density is instead defined as

$$D_{\mathrm{dir}}(G) = \frac{M_{\mathrm{dir}}}{N(N-1)}, \tag{24}$$

where M_{dir} denotes the number of arcs.

These indicators provide a description of the relative abundance of connections within the school.

However, their interpretation must take into account the number of observed individuals and the selected neighbourhood radius.

3.6 Graph Distances, Diameter and Radius

The distance between two vertices is defined as the length of the shortest path connecting them.

In the present analysis, the path length is expressed in numbers of edges, or hops, rather than in physical distance units.

Let

$$d_G(i,j) \tag{25}$$

denote the shortest-path distance between vertices i and j.

The eccentricity of a vertex is defined as

$$e(i) = \max_{j \in V} d_G(i,j). \tag{26}$$

The diameter of a connected graph is then

$$\mathrm{diam}(G) = \max_{i \in V} e(i), \tag{27}$$

while its radius is

$$\mathrm{rad}(G) = \min_{i \in V} e(i). \tag{28}$$

Since the observed networks may contain several components, these measures cannot be interpreted without specifying the connected subgraph considered.

For the spatial model, the diameter and radius are calculated on the largest connected component.

For the directed visibility model, they are calculated on the largest strongly connected component.

This convention avoids assigning finite distances between vertices for which no path exists.

3.7 Centrality Measures

Centrality measures are used to identify individuals occupying structurally prominent positions within the network. Two measures are retained in the principal analysis.

3.7.1 Degree Centrality

For an undirected graph, degree centrality is defined by

$$C_D(i) = \frac{k_i}{N-1}. \tag{29}$$

It expresses the proportion of other individuals directly connected to vertex i.

A high degree centrality may indicate that a fish is situated in a locally dense region of the school.

For the directed visibility graph, in-degree and out-degree centralities are distinguished.

The out-degree counts the individuals towards which a fish has an outgoing visibility connection, whereas the in-degree counts the incoming visibility connections.

3.7.2 Betweenness Centrality

Betweenness centrality measures the extent to which a vertex belongs to shortest paths connecting other vertices. It is defined by

$$C_B(i) = \sum_{\substack{s \neq i \neq t \\ s \neq t}} \frac{\sigma_{st}(i)}{\sigma_{st}}, \tag{30}$$

where σ_{st} is the number of shortest paths between vertices s and t, and $\sigma_{st}(i)$ the number of those paths passing through vertex i.

The values are normalised using the convention implemented in NetworkX.

An individual with high betweenness centrality may occupy a position connecting different regions of the graph. However, such a position does not establish that the individual controls the movements of the school or acts as a biological leader.

3.8 Clustering Coefficient

The clustering coefficient describes the tendency of neighbouring vertices to be connected to one another.

For an undirected graph, the local clustering coefficient of vertex i is

$$C_i = \frac{2T_i}{k_i(k_i-1)}, \tag{31}$$

where T_i denotes the number of triangles containing vertex i.

The average clustering coefficient is

$$\overline{C} = \frac{1}{N}\sum_{i=1}^{N} C_i. \tag{32}$$

A relatively high clustering coefficient indicates that neighbouring fish frequently belong to interconnected local groups.

For directed graphs, the NetworkX implementation uses a corresponding definition adapted to the orientation of the arcs.

3.9 Community Detection

The identification of communities provides an additional description of the internal organisation of the fish school.

A community is understood here as a subset of vertices having relatively dense internal connections compared with their connections to other subsets.

Community detection is performed using the Louvain algorithm, as implemented in NetworkX. The algorithm seeks a partition associated with relatively high modularity.

3.9.0.1 Modularity of the undirected spatial graph. For the undirected spatial proximity graph, modularity is defined as

$$Q_{\text{undirected}} = \frac{1}{2M}\sum_{i,j}\left(A_{ij} - \frac{k_i k_j}{2M}\right)\delta(c_i, c_j), \tag{33}$$

where A_{ij} is the adjacency matrix, k_i and k_j are the degrees of vertices i and j, and M is the number of undirected edges.

The term $\delta(c_i, c_j)$ equals one when vertices i and j belong to the same community, and zero otherwise.

3.9.0.2 Modularity of the directed visibility graph. The visibility graph is directed because the visibility relationship between two fish is not necessarily reciprocal.

Louvain community detection is applied directly to this directed graph, without converting it into an undirected representation.

For a directed graph, modularity is defined as

$$Q_{\text{directed}} = \frac{1}{M} \sum_{i,j} \left(A_{ij} - \frac{k_i^{\text{out}} k_j^{\text{in}}}{M} \right) \delta(c_i, c_j), \tag{34}$$

where M is the total number of directed arcs, k_i^{out} is the out-degree of vertex i, and k_j^{in} is the in-degree of vertex j.

This formulation accounts for the orientation of interactions and is consistent with the directed modularity implemented in NetworkX.

3.9.0.3 Implementation and parameters. Both graph models are analysed using the following parameters :

— Algorithm : Louvain (NetworkX) ;
— Random seed : 42 ;
— Resolution parameter : 1.0 ;
— Edge weights : not used ;
— Spatial graph : undirected ;
— Visibility graph : directed.

The resulting partitions are evaluated using the number of detected communities, their sizes, and the corresponding modularity.

For the first experimental frame, the spatial graph produces 11 communities with a modularity of 0.70344, while the directed visibility graph produces 10 communities with a modularity of 0.70500.

These results are consistent with the outputs stored in the project repository.

Although both modularity values are numerically close, they are calculated using different null models. Their comparison must therefore be interpreted with caution.

Furthermore, communities identified in spatial proximity graphs may arise partly from their geometrical construction.

They must not automatically be interpreted as stable biological groups or as evidence of coordinated decision-making.

3.10 Temporal Analysis and Sensitivity

The dataset contains 5,000 observation frames, recorded at 30 frames per second.

In order to investigate variations in network organisation, a systematic sample of 21 frames is considered :

$$t \in \{1, 251, 501, \ldots, 4751, 5000\}. \tag{35}$$

The corresponding observations cover approximately 166.63 seconds.

For each selected frame, the two graph models are constructed and their principal structural properties are calculated.

The analysis is repeated for five neighbourhood radii :

$$R \in \{60, 80, 100, 120, 140\}. \tag{36}$$

This approach makes it possible to distinguish variations associated with the observed spatial configurations from those introduced by the graph construction threshold.

The results are summarised using descriptive statistics and graphical representations.

The temporal curves connect sampled observations ; they do not establish the behaviour of the network between successive sampled frames.

4 Results

4.1 Structural Properties of the First Frame

The first analysis is performed on frame 1, using a neighbourhood radius of $R = 100$.

The experimental data contain 253 detected fish at this observation time.

The spatial graph contains 1,281 undirected edges, while the directed visibility graph contains 2,386 arcs.

The principal results are presented in Table 1.

The spatial graph contains four connected components, with the largest component comprising 248 of the 253 observed individuals.

This corresponds to approximately 98.02% of the vertices.

The results therefore indicate that most of the observed fish belong to a common connected structure under the selected proximity criterion.

Nevertheless, the existence of four components shows that the graph is not entirely connected.

TABLE 1 – Structural properties of the two graph models at frame 1, for $R = 100$.

Measure	Spatial	Visibility
Vertices	253	253
Edges / arcs	1,281	2,386
Relevant components	4	4
Largest component	248	248
Largest component share	98.02%	98.02%
Diameter (hops)	23	24
Radius (hops)	12	12
Density	0.04018	0.03742
Average clustering	0.57800	0.54551
Louvain communities	11	10
Modularity	0.70344	0.70500

The diameter of the largest spatial component is 23 hops, while its radius is 12 hops.

These values describe the shortest-path structure of the network and should not be confused with the Euclidean distances between fish.

The directed visibility graph presents a comparable global structure, although the directional constraint modifies the number and orientation of the available connections.

Its largest strongly connected component contains 248 vertices, with a diameter of 24 hops and a radius of 12 hops.

4.2 Centrality and Structural Positions

The degree and betweenness centralities reveal differences in the structural positions of individual fish.

In the spatial graph, the highest degree centrality is associated with fish identifier 100230, with a normalised score of 0.09921.

The highest betweenness centrality is observed for fish identifier 99458, with a score of 0.16501.

These two individuals do not occupy the same position in the centrality rankings.

This distinction illustrates the complementary interpretations provided by the two measures.

A fish may be connected to numerous immediate neighbours without necessarily occupying a position through which a large proportion of shortest paths passes.

Conversely, an individual may connect different regions of the graph while having a more moderate number of direct neighbours.

For the visibility graph, the highest betweenness centrality is associated with identifier 99981, with a score of 0.14945.

The introduction of directional visibility therefore modifies the ranking of structurally prominent individuals.

These observations concern network positions only. They do not demonstrate the existence of behavioural leadership.

4.3 Community Structure

Community detection provides a complementary perspective on the internal organisation of the fish school. While connectivity measures establish whether individuals can be connected through paths, community detection seeks to identify groups of vertices exhibiting comparatively dense internal connections.

The Louvain algorithm was used to identify community partitions in the spatial and visibility networks. For the first experimental frame, the spatial graph contains 253 observed fish and 1,281 undirected edges, whereas the visibility graph contains the same 253 individuals and 2,386 directed arcs.

The resulting partitions exhibit substantial modularity. The spatial graph contains 11 detected communities, with a modularity of

$$Q_{\text{spatial}} = 0.70344. \tag{37}$$

The visibility graph contains 10 communities, with

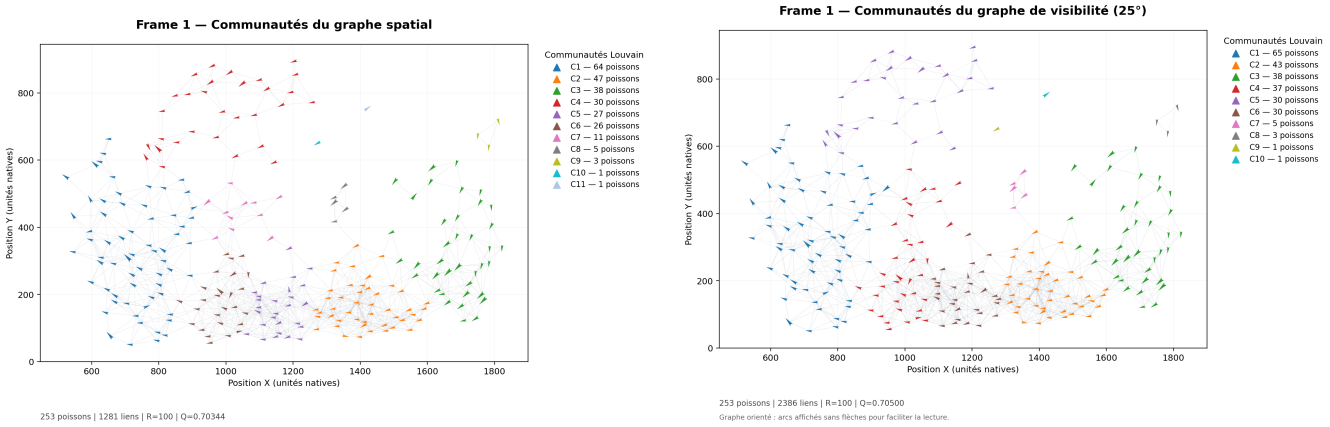
$$Q_{\text{visibility}} = 0.70500. \tag{38}$$

The largest community of the spatial graph contains 64 individuals, followed by a second community containing 47 fish. In the visibility graph, the two largest communities contain 65 and 43 individuals, respectively.

Figure 1 illustrates the geographical distribution of these partitions.

The visualisations reveal a pronounced spatial organisation of the detected communities. Several groups occupy geographically concentrated regions of the school, whereas the central region contains individuals assigned to different communities.

The introduction of the directional visibility constraint modifies the distribution of individuals among the detected groups. Nevertheless, the modularity values remain remarkably close, indicating that the overall strength of the detected community structure is similar under the two representations.



(a) Spatial proximity network.

(b) Directed visibility network.

FIGURE 1 – Louvain community partitions in the first experimental frame. Both networks contain 253 fish, with a neighbourhood radius of $R = 100$ and a rear blind angle of 25° for the visibility model. Colours identify independently detected communities in each graph. The directed arcs of the visibility network are represented without arrowheads to improve readability.

An important distinction must be established between connected components and communities. A graph may contain a single dominant connected component while still exhibiting several internally differentiated communities. These two properties therefore describe different aspects of network organisation.

However, the interpretation of the detected communities requires particular caution. Since connections are constructed from spatial proximity and directional visibility, the geometrical configuration of the fish school may itself generate relatively dense substructures.

Consequently, the community partitions should be interpreted as topological descriptions of the observed network rather than direct evidence of stable biological or social subgroups.

4.4 Temporal Variations in Network Organisation

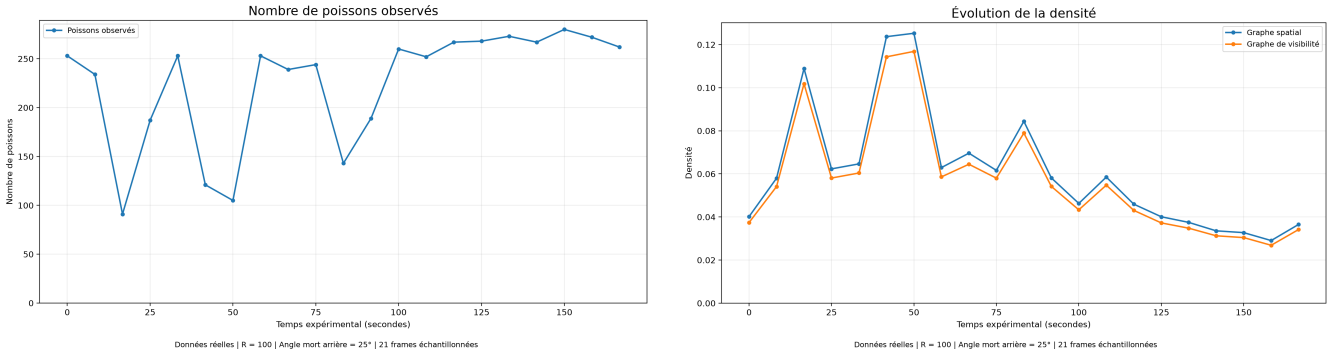
The analysis of a single experimental frame provides a detailed description of the network at a particular instant. However, fish schooling is a dynamic phenomenon, and its structural organisation may evolve throughout the recording.

To investigate these variations, 21 experimental frames were selected across the observation period. For each frame, both graph representations were constructed using the same neighbourhood radius of $R = 100$ and a rear blind angle of 25°.

The temporal analysis considers the number of observed individuals, graph density, connectivity, clustering and community modularity.

4.4.0.1 Number of observed fish and network density The number of observed individuals varies considerably throughout the sampled sequence. The smallest observation contains 91 fish, whereas the largest contains 280 individuals.

Figure 2 presents these variations together with the corresponding evolution of graph density.



(a) Number of observed fish.

(b) Evolution of graph density.

FIGURE 2 – Temporal evolution of the number of observed fish and the density of the spatial and visibility networks across 21 sampled frames. The neighbourhood radius is $R = 100$, with a rear blind angle of 25° for the visibility network.

The first part of the recording exhibits substantial fluctuations in the number of detected individuals, followed by comparatively higher and more stable observations towards the end.

The spatial graph density ranges from approximately 0.029 to 0.125. The visibility graph follows a similar temporal trajectory, although its density remains consistently lower.

This difference is consistent with the additional directional restriction introduced by the visibility model.

Nevertheless, the temporal variations in graph density cannot be interpreted independently of the observed population. Changes in the number of detected fish modify the number of possible connections and may influence the measured topology.

Consequently, changes in network density cannot be attributed exclusively to modifications in collective behaviour.

4.4.0.2 Connectivity of the principal components The temporal evolution of connectivity is examined through the relative size of the largest connected components.

For the spatial graph, the largest connected component is considered. For the directed visibility graph, the largest strongly connected component is used.

Figure 3 presents the corresponding results.

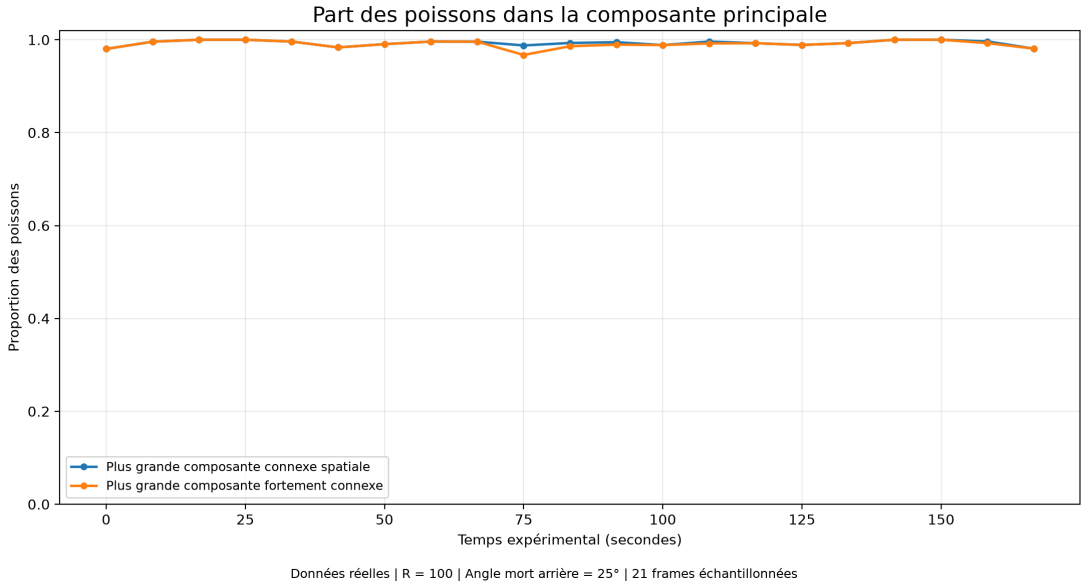


FIGURE 3 – Temporal evolution of the proportion of observed fish belonging to the largest connected component of the spatial graph and the largest strongly connected component of the directed visibility graph.

The spatial graph exhibits a particularly high level of global connectivity. The proportion of fish belonging to its largest connected component remains between approximately 98% and 100%.

The visibility graph displays a closely related pattern, with the largest strongly connected component containing almost all observed individuals.

These results indicate that the principal network structure remains largely connected throughout the sampled sequence, despite variations in the observed population and network density.

However, the persistence of a dominant connected component does not imply that the same individuals maintain identical relationships throughout the experiment.

Connectivity describes the existence of paths within each observed network, rather than the temporal persistence of individual connections.

4.4.0.3 Clustering and community modularity The evolution of local clustering and community modularity provides additional information about the internal organisation of the fish school.

Figure 4 compares these measures for the spatial and visibility networks.

The average clustering coefficient remains relatively high throughout the experimental sequence. For the spatial graph, it varies between approximately 0.53 and 0.63.

The visibility graph generally exhibits slightly lower values while preserving a similar temporal evolution.

These results indicate that neighbouring individuals frequently form locally interconnected groups.

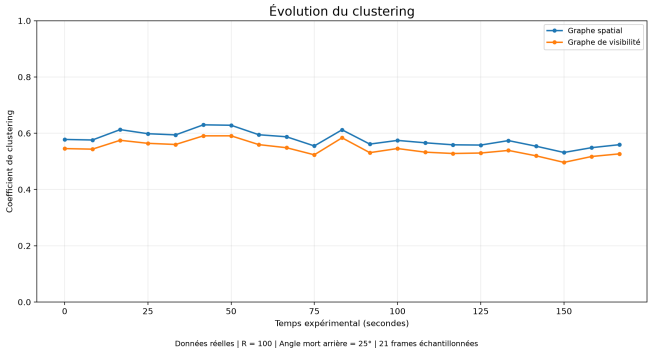
The Louvain modularity also varies throughout the observation period. For the spatial graph, values range from approximately 0.57 to 0.73.

The two models follow closely related modularity trajectories, suggesting that the additional directional constraint does not substantially modify the overall temporal pattern of community organisation under the selected parameters.

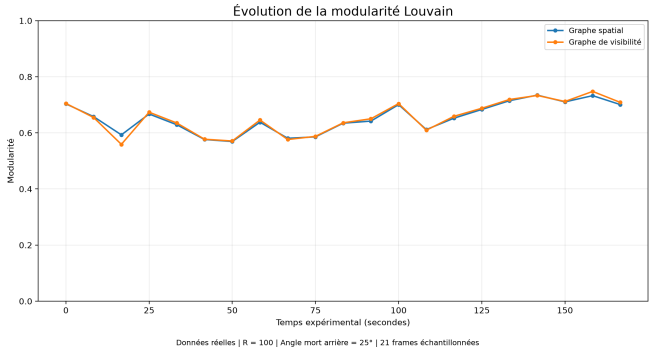
Nevertheless, variations in modularity may result from changes in spatial configuration, the number of detected individuals or the community partitions identified by the algorithm.

The temporal analysis therefore reveals a network structure characterised by high global connectivity, relatively pronounced local clustering and persistent modular organisation at the level of the sampled graphs.

These observations provide evidence of structural regularities in the constructed networks, but do not establish the existence of fixed social relationships or causal interaction mechanisms.



(a) Average clustering coefficient.



(b) Louvain modularity.

FIGURE 4 – Temporal evolution of the average clustering coefficient and Louvain modularity for both network representations, evaluated across the same 21 sampled frames.

4.5 Sensitivity to the Neighbourhood Radius

The construction of a spatial proximity graph requires the selection of a neighbourhood radius. This parameter determines which individuals are considered sufficiently close to establish a connection.

Since the resulting topology depends directly on this modelling choice, a sensitivity analysis was conducted to evaluate its influence on the main structural measures.

Five neighbourhood radii were considered :

$$R \in \{60, 80, 100, 120, 140\}. \tag{39}$$

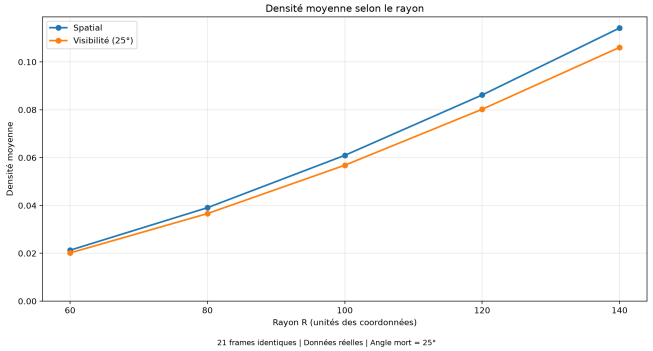
The analysis was performed using the same 21 experimental frames for each radius. The rear blind angle of the visibility graph was maintained at 25°.

Table 2 summarises the mean structural properties obtained for the spatial graph.

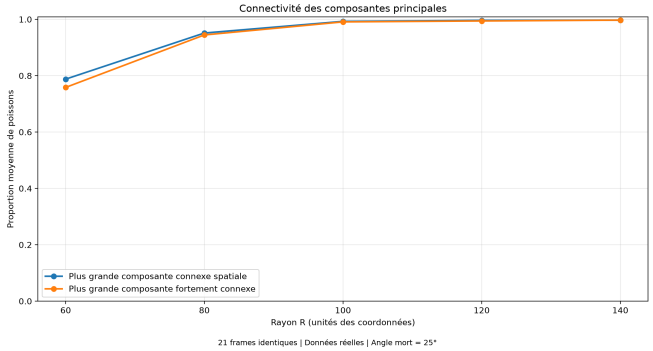
TABLE 2 – Mean structural properties of the spatial graph across 21 experimental frames for different neighbourhood radii.

R	Density	Clustering	Communities	LCC (%)
60	0.02128	0.41344	29.76	78.78
80	0.03906	0.52683	12.90	95.14
100	0.06095	0.57873	8.76	99.28
120	0.08619	0.61020	6.81	99.63
140	0.11418	0.63357	6.43	99.71

4.5.0.1 Network density and connectivity Figure 5 presents the evolution of network density and the relative size of the principal connected components as the neighbourhood radius increases.



(a) Mean network density.



(b) Largest component share.

FIGURE 5 – Influence of the neighbourhood radius on network density and the relative size of the largest connected components. The same 21 frames are used for both graph representations.

The mean density increases systematically with the neighbourhood radius.

For the spatial graph, it rises from 0.02128 at $R = 60$ to 0.11418 at $R = 140$.

The visibility graph exhibits a similar evolution, although its density remains slightly lower for every selected radius.

This behaviour follows directly from the graph construction procedure. Increasing the neighbourhood radius introduces additional potential connections between individuals whose positions remain unchanged.

The relative size of the largest connected component also increases with the neighbourhood radius.

At $R = 60$, the largest connected component of the spatial graph contains approximately 78.78% of the observed individuals.

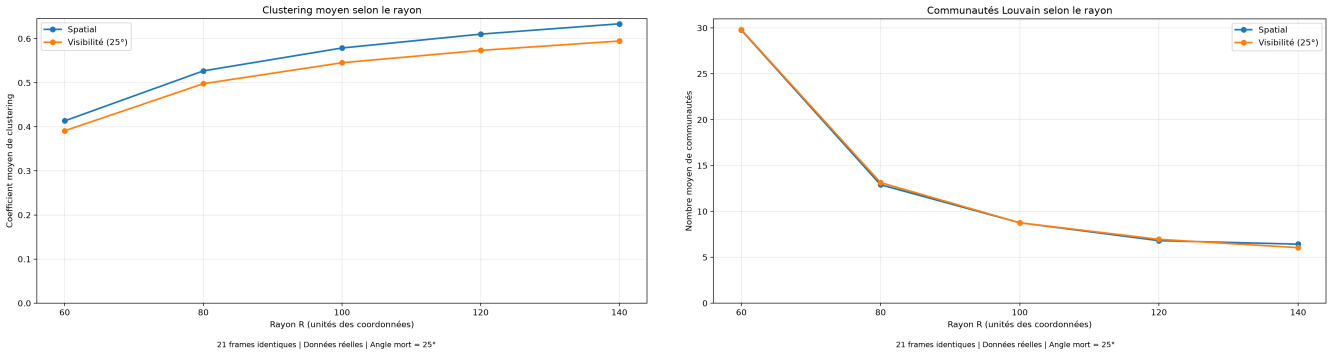
This proportion increases to 95.14% at $R = 80$ and 99.28% at $R = 100$.

Beyond $R = 100$, the largest component already contains almost all observed fish, and further increases in radius produce relatively limited changes in global connectivity.

The visibility graph exhibits a comparable tendency, although its connectivity is evaluated through the largest strongly connected component.

These results indicate that the neighbourhood radius has a particularly pronounced influence on network fragmentation at smaller values.

4.5.0.2 Clustering and community organisation The effect of the neighbourhood radius on local clustering and the number of detected communities is presented in Figure 6.



(a) Mean clustering coefficient.

(b) Mean number of communities.

FIGURE 6 – Evolution of the mean clustering coefficient and the number of Louvain communities as a function of the neighbourhood radius.

The mean clustering coefficient increases from 0.41344 at $R = 60$ to 0.63357 at $R = 140$ in the spatial graph.

The visibility graph follows a similar trajectory, with slightly lower clustering values.

This increase indicates that larger neighbourhood radii produce a greater proportion of interconnected neighbours, resulting in denser local structures.

Conversely, the mean number of detected communities decreases as the neighbourhood radius increases.

For the spatial graph, the mean number of communities declines from 29.76 at $R = 60$ to 6.43 at $R = 140$.

A comparable evolution is observed for the visibility graph.

At smaller radii, the network contains numerous local groups, some of which are disconnected or weakly connected to the remaining structure.

As the radius increases, additional connections emerge between these groups, reducing the degree of network fragmentation.

The simultaneous increase in clustering and decrease in the number of communities illustrates the substantial influence of the interaction radius on the organisation of the constructed network.

4.5.0.3 Sensitivity of Louvain modularity Figure 7 illustrates the evolution of mean Louvain modularity for the five selected neighbourhood radii.

The mean modularity decreases as the neighbourhood radius increases.

For both graph representations, modularity is approximately 0.80 at $R = 60$ and decreases to approximately 0.56 at $R = 140$.

The two curves remain particularly close throughout the analysis, indicating that the overall trend is comparable between the spatial and visibility models.

This decrease suggests that the relative distinction between detected communities becomes less pronounced as the network becomes increasingly connected.

At smaller radii, the graph contains more clearly separated local structures, producing higher modularity values.

At larger radii, additional connections between previously distinct groups reduce their relative separation.

Nevertheless, the modularity values depend on the network construction rule and the community detection procedure.

They must therefore be interpreted as structural properties of the selected graph representations, rather than direct measurements of biological social organisation.

4.5.0.4 Interpretation of the reference radius The sensitivity analysis demonstrates that the neighbourhood radius substantially influences the topological properties of the constructed networks.

A smaller radius produces sparser and more fragmented graphs, while a larger radius generates denser structures with stronger global connectivity and fewer detected communities.

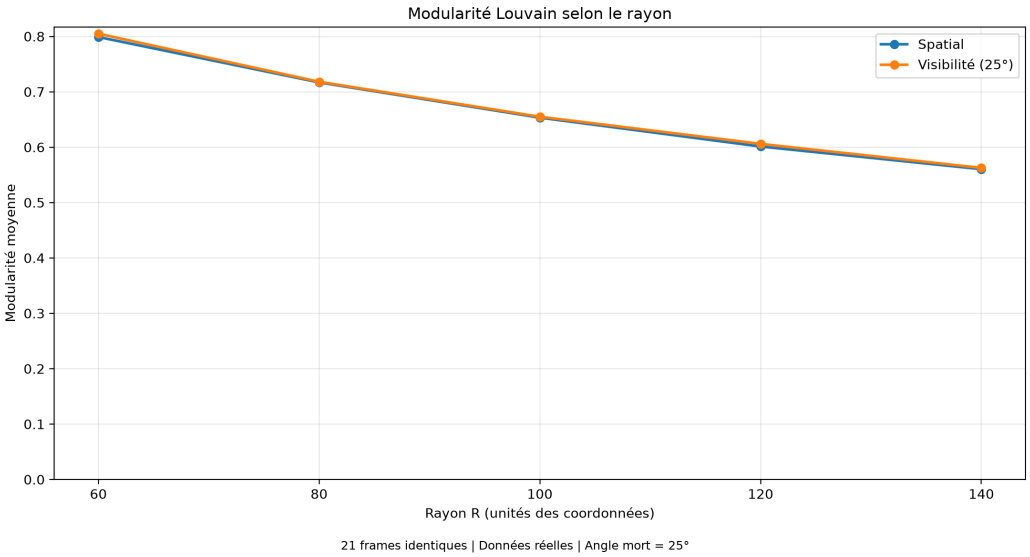


FIGURE 7 – Mean Louvain modularity as a function of the neighbourhood radius for the spatial and visibility networks, evaluated across 21 identical experimental frames.

The reference value $R = 100$ provides a practical compromise for the present analysis.

At this radius, the largest connected component contains more than 99% of the observed individuals on average, while the network retains a differentiated community structure.

However, this choice remains a modelling assumption.

The present results do not establish that $R = 100$ corresponds to an optimal biological interaction distance.

The sensitivity analysis therefore provides an essential means of distinguishing relatively robust structural observations from properties induced by the selected neighbourhood threshold.

4.6 Comparison of Spatial and Visibility Models

At $R = 100$, the two models exhibit similar average structural properties across the 21 selected frames.

The mean spatial density is 0.06095, compared with 0.05682 for the directed visibility graph.

The corresponding mean clustering coefficients are 0.57873 and 0.54529.

The mean modularity values are 0.65325 and 0.65490, respectively.

The directional visibility constraint therefore reduces the number of available connections without producing a complete reorganisation of the network under the selected angular parameter.

On average, the visibility graph retains approximately 93.22% of the potential directed connections between spatially neighbouring fish.

This result is consistent with the relatively narrow rear blind sector considered in the model.

Nevertheless, the comparison must take account of the differences between directed and undirected network measures.

In particular, the largest connected component of the spatial graph and the largest strongly connected component of the visibility graph do not represent identical accessibility conditions.

5 Conclusion

The analysis of the fish school through spatial proximity and directed visibility graphs provides a mathematical description of its collective organisation.

By representing each observed individual as a vertex and defining connections according to spatial and directional criteria, it becomes possible to examine the connectivity, distances, centrality and community structure of the school.

The results obtained for the first observation frame show that most individuals belong to a common connected structure.

At the same time, the identification of several communities indicates that a globally connected network may contain relatively distinct regions of local organisation.

The analysis of centrality measures also reveals differences between individuals having numerous direct connections and those occupying positions of relatively high betweenness.

These differences provide useful information about the topology of the network, although they cannot establish the existence of behavioural leadership.

The temporal analysis indicates that the structural properties of the observed networks vary throughout the experimental sequence.

However, the interpretation of these variations must take into account changes in the number of detected individuals and the dependence of the graph on its construction parameters.

The sensitivity analysis confirms the importance of the neighbourhood radius.

Increasing this parameter produces denser graphs, larger connected components and a reduction in the mean number of detected communities.

The reference radius of 100 native coordinate units therefore provides a useful basis for descriptive comparisons, without constituting a biologically validated threshold.

The directed visibility model introduces an additional level of representation by considering the approximate orientation of individual movements.

Under the selected rear blind angle, its structural properties remain relatively close to those of the spatial graph.

The principal limitation of the study lies in the distinction between spatial proximity and actual behavioural interactions.

The existence of a connection between two individuals does not demonstrate communication, influence or causal dependence.

Likewise, the community structure identified by the algorithms may partly reflect the geometrical properties of the graph construction.

The present work therefore demonstrates the relevance of graph-theoretical methods for describing the structural organisation of a fish school, while emphasising the need for caution when translating network measures into biological interpretations.

Further investigations may consider the temporal persistence of individual connections, alternative neighbourhood definitions and comparisons between observed trajectories and simulated collective behaviours.

5.0.0.1 Code Availability and Reproducibility. The source code, graph construction and analysis scripts, Boids simulation implementations, and available experimental results are accessible through the project's public GitHub repository :

`https://github.com/Charles-Scannavino/Projet_Graph_Analytics`

The repository provides the implementation details required to examine the graph construction methods, network measures, community detection procedures, and simulation experiments presented in this report.

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Annexe A – Fish-School Modelling and Objectives

Annexe A.1 Existing Models of the Problem

This problem has given rise to several types of modelling.

One of the most interesting lines of research is associated with Giorgio Parisi and his team, whose work on complex systems was recognised by the 2021 Nobel Prize in Physics.

The model considered here is based on an earlier approach, which remains highly influential in the literature on **collective intelligence and the dynamics of distributed agents** : that of Craig Reynolds (1987).

Annexe A.2 Continuous Model of Propagation and Cohesion–Repulsion in a Fish School

Modelling of Individuals

We consider a set of N fish, each identified by an index $i \in \{1, \dots, N\}$, and characterised by a position $X_i(t) \in \mathbb{R}^2$ and a velocity $V_i(t) = \dot{X}_i(t)$.

The direction of movement is given by the angle

$$D_i(t) = \operatorname{atan2}(v_i^y, v_i^x),$$

so that

$$V_i(t) = v_i\big(\cos D_i(t), \sin D_i(t)\big).$$

The collective behaviour of the school is described by the dynamics

$$\frac{dV_i}{dt} = F_{\text{cohesion},i} + F_{\text{repulsion},i} + F_{\text{alignment},i} + F_{\text{danger},i},$$

where each term represents a local interaction.

The Three Fundamental Forces

The collective motion of the fish school is represented through three local interaction mechanisms : cohesion, repulsion (separation), and alignment.

Each mechanism operates within a potentially different interaction radius.

For fish i, the corresponding neighbourhoods are defined as

$$\mathcal{N}_i^{(c)} = \{j \neq i : \|X_j - X_i\| < R_c\},$$

$$\mathcal{N}_i^{(s)} = \{j \neq i : \|X_j - X_i\| < R_s\},$$

$$\mathcal{N}_i^{(a)} = \{j \neq i : \|X_j - X_i\| < R_a\}.$$

Here, R_c, R_s, and R_a denote the cohesion, separation (repulsion), and alignment radii, respectively.

The separation radius is assumed to be smaller than the cohesion radius :

$$R_s < R_c.$$

No additional ordering between the alignment radius and the other radii is required by the theoretical formulation.

Annexe A.2.0.1 a) Cohesion. Each fish is attracted towards the local centre of mass of neighbouring individuals.

The cohesion force is defined as

$$F_{\text{cohesion},i} = k_c \left(\bar{X}_i^{(c)} - X_i\right),$$

where

$$\bar{X}_i^{(c)} = \frac{1}{|\mathcal{N}_i^{(c)}|} \sum_{j \in \mathcal{N}_i^{(c)}} X_j.$$

The parameter $k_c > 0$ controls the intensity of the cohesion force.

If the cohesion neighbourhood is empty, the cohesion force is set to zero.

Annexe A.2.0.2 b) Repulsion (Separation). A short-range repulsive interaction prevents fish from approaching one another too closely.

The repulsion force is defined as

$$F_{\text{repulsion},i} = -k_r \sum_{j \in \mathcal{N}_i^{(s)}} \frac{X_j - X_i}{\|X_j - X_i\|^2 + \varepsilon},$$

where $k_r > 0$ controls the strength of repulsion and $\varepsilon > 0$ prevents division by zero.

The negative sign ensures that the force points away from nearby individuals.

Only fish belonging to the separation neighbourhood contribute to this force.

If no neighbour is present within R_s, the repulsion force is zero.

Annexe A.2.0.3 c) Alignment. Each fish tends to adjust its velocity towards the average velocity of its alignment neighbours.

The alignment force is defined as

$$F_{\text{alignment},i} = k_a \left(\bar{V}_i^{(a)} - V_i \right),$$

where

$$\bar{V}_i^{(a)} = \frac{1}{|\mathcal{N}_i^{(a)}|} \sum_{j \in \mathcal{N}_i^{(a)}} V_j.$$

The parameter $k_a > 0$ controls the intensity of velocity adjustment.

If the alignment neighbourhood is empty, the alignment force is set to zero.

Annexe A.2.0.4 Combined dynamics. The three interactions contribute to the continuous-time velocity dynamics :

$$\frac{dV_i}{dt} = F_{\text{cohesion},i} + F_{\text{repulsion},i} + F_{\text{alignment},i} + F_{\text{danger},i}.$$

The danger term represents an optional external perturbation, described separately in the following subsection.

The equations above provide a theoretical description of collective motion. The discrete-time Boids implementation used in the experiments is described in Appendix B.

Danger Force and Directional Propagation

A localised danger, such as a predator, obstacle, or sudden noise, may be detected by a fish located at the edge of the school, identified by the index i_0.

As an illustrative scenario, the detecting fish may initially change its direction by 45° in response to the threat. This angular change is a prescribed behavioural assumption ; it is not directly imposed by the repulsive force below.

Let X_{danger} denote the position of the danger source and X_{i_0} the position of the detecting fish. The repulsive force is defined as

$$F_{\text{danger},i_0} = -k_d \frac{X_{\text{danger}} - X_{i_0}}{\|X_{\text{danger}} - X_{i_0}\| + \varepsilon}, \qquad k_d > 0, \quad \varepsilon > 0.$$

The negative sign ensures that the force points away from the danger source. The parameter k_d controls its strength, while ε prevents division by zero.

Annexe A.2.0.5 Propagation of the directional response. The initial change in direction may subsequently propagate through neighbouring fish. A simplified continuous model is

$$\frac{dD_i}{dt} = \lambda \ \text{wrap}(D_{i-1} - D_i), \qquad \lambda > 0,$$

where D_i denotes the direction of fish i and D_{i-1} that of a fish responding earlier.

The index $i-1$ refers to a predecessor in an assumed propagation sequence, not necessarily the preceding tracking identifier in the dataset.

The operator wrap returns the shortest signed angular difference in $[-\pi, \pi]$, and λ controls the adjustment rate.

A corresponding discrete-time directional adjustment is

$$D_i^{(t+1)} = D_i^{(t)} + \alpha \ \text{wrap}\left(D_{i-1}^{(t)} - D_i^{(t)} \right), \qquad 0 < \alpha < 1.$$

This formulation illustrates how a local directional change could spread through a school.

It is a theoretical mechanism, not an experimentally validated model of escape-wave transmission in the observed dataset.

Annexe A.3 Further Details on the Equations of Motion

Annexe A.3.1 Complete Equation of Motion

The overall dynamics are written as

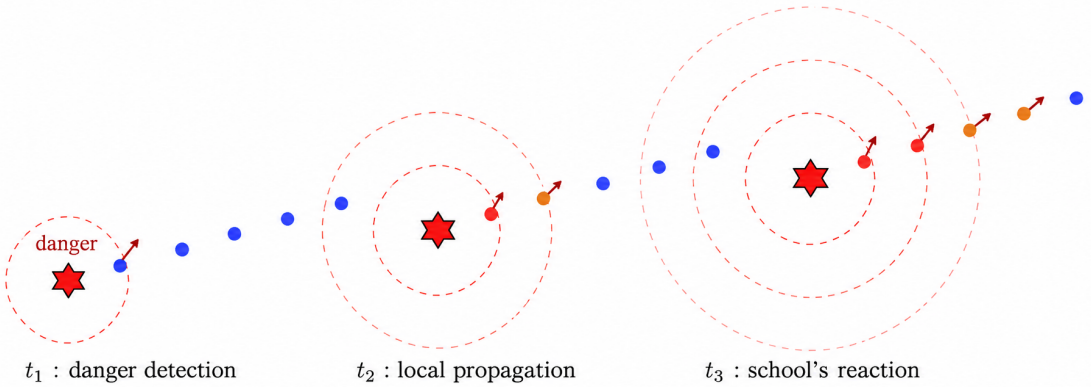


FIGURE A.1 – Temporal propagation of the escape wave in a *fish* school (three successive stages).

$$\begin{aligned}\frac{dX_i}{dt} &= V_i,\\ \frac{dV_i}{dt} &= k_c(\bar{X}_i - X_i) - k_r \sum_{j\in\mathcal{N}_i} \frac{X_j - X_i}{\|X_j - X_i\|^2 + \varepsilon}\\ &\quad + k_a(\bar{V}_i - V_i) + F_{\text{danger},i}.\end{aligned}$$

This nonlinear coupled system is of the *agent-based continuous system* type.

Depending on its parameters and initial conditions, it can produce collective motion, dispersion, re-aggregation and changes in group organisation.

Annexe A.3.2 Cohesion and Repulsion Mechanisms

Interaction	Range	Formula	Overall effect
Cohesion	$r < R_c$	$k_c(\bar{X}_i - X_i)$	Group maintenance
Repulsion	$r < R_s < R_c$	$-k_r(X_j - X_i)/r^2$	Collision avoidance
Alignment	$r < R_a$	$k_a(\bar{V}_i - V_i)$	Collective motion
Danger	Localised	$-k_d(X_{\text{danger}} - X_i)/r$	Escape response

TABLE 3 – Local interactions, ranges of action and their principal collective effects. The simplified expressions in the table omit regularisation terms.

These interactions are linearly superimposed and integrated into the continuous dynamics, giving rise to different forms of collective behaviour.

Annexe A.4 Project Objectives

Collective intelligence and the dynamics of distributed agents concern the ability of a set of agents—individuals, animals, robots, etc.— to produce coherent behaviour adapted to certain situations in a decentralised manner, without requiring a leader or central control.

Several hypotheses can therefore be considered to obtain collective behaviours through dynamic equations that can serve as a basis for simulation.

The resulting evolutions can then be compared with collective movements recorded in a dataset.

The objective is to determine to what extent local behavioural rules can reproduce the spatial and structural organisation of an experimentally observed fish school.

Annexe B – The Boids Model of Craig Reynolds (1987)

Annexe B.1 General Introduction

The *boids* model (a portmanteau formed from *bird-oid object*) was proposed in 1987 by Craig W. Reynolds in an article that has become a classic :

> Craig W. Reynolds, "Flocks, Herds, and Schools : A Distributed Behavioral Model", *Computer Graphics (SIGGRAPH '87 Conference Proceedings)*, Vol. 21, No. 4, pp. 25–34, 1987.

It is a computational model of collective behaviour, initially designed to simulate the flight of birds or the movement of fish schools in three-dimensional space.

This model is based on the idea that group coherence does not require central coordination : it emerges from simple local rules followed by each agent.

Annexe B.2 Fundamental Principle : Local Behaviour, Global Emergence

Each agent (or *boid*) moves in a continuous space, subject to three local behavioural rules :

1. **Separation** : avoid collisions with immediate neighbours ;
2. **Alignment** : move in the same direction as nearby neighbours ;
3. **Cohesion** : move closer to the local centre of gravity of the group.

Without any global command, these rules can produce emergent collective behaviour : swimming in formation, synchronised changes in direction and the dynamic organisation of a school.

Annexe B.3 Mathematical Formalisation

Let N be the number of agents, each defined by its position $X_i(t) \in \mathbb{R}^d$ and velocity $V_i(t) = \dot{X}_i(t)$, with $d = 2$ or 3.

The temporal evolution of the system follows the equation

$$\frac{dV_i}{dt} = F_{\text{separation},i} + F_{\text{alignment},i} + F_{\text{cohesion},i} + F_{\text{other}},$$

where F_{other} may include additional forces such as obstacles, predators or goals to be reached.

Annexe B.3.0.1 1. Separation (or repulsion).

$$F_{\text{separation},i} = -k_s \sum_{j \in \mathcal{N}_i^{(s)}} \frac{X_j - X_i}{\|X_j - X_i\|^2 + \varepsilon}.$$

Here, $\mathcal{N}_i^{(s)}$ is the set of neighbours at a distance smaller than R_s.

This force pushes boid i away from any neighbour that is too close, preventing collisions and maintaining a minimum distance.

Annexe B.3.0.2 2. Alignment.

$$F_{\text{alignment},i} = k_a(\bar{V}_i - V_i), \qquad \bar{V}_i = \frac{1}{|\mathcal{N}_i^{(a)}|} \sum_{j \in \mathcal{N}_i^{(a)}} V_j.$$

Here, $\mathcal{N}_i^{(a)}$ denotes the neighbours within a radius R_a.

The boid adjusts its velocity to the average local velocity.

Annexe B.3.0.3 3. Cohesion (or attraction).

$$F_{\text{cohesion},i} = k_c(\bar{X}_i - X_i), \qquad \bar{X}_i = \frac{1}{|\mathcal{N}_i^{(c)}|} \sum_{j \in \mathcal{N}_i^{(c)}} X_j.$$

This force attracts each boid towards the centre of gravity of its neighbours within a radius R_c.

Annexe B.4 Mechanism of Emergence

The three local rules, applied simultaneously to each agent, can generate coherent and dynamic collective structures :

— Separation prevents internal collisions.
— Alignment promotes directional coherence.
— Cohesion maintains group compactness.

The global behaviours observed—formation, avoidance and collective rotation—are *emergent phenomena*, rather than behaviours explicitly specified in the model.

This approach illustrates the idea that *simple local rules* can generate *complex global patterns*.

Annexe B.5 Extensions of the Model

Since 1987, the boids model has undergone numerous extensions :

1. **Boids with limited perception** : introduction of a field of vision restricting interactions to visible neighbours.
2. **Boids with inertia** : addition of inertial terms to velocities to simulate the physical continuity of movement.
3. **Hierarchical boids** : leader or dominant agents orienting the group.
4. **Environmental boids** : consideration of obstacles, currents or external potential fields.
5. **Emotional or cognitive boids** : addition of internal variables such as stress, memory or goals.

These variations make it possible to adapt the model to different contexts : biology, swarm robotics, three-dimensional animation and crowd simulation.

Annexe B.6 Relationship with Physical and Biological Models

The Reynolds model, although simple, is related to several theoretical frameworks :

— the dynamics of continuous multi-agent systems ;
— mean-field equations used in statistical physics ;
— biological models of local interaction (Cucker–Smale, Vicsek, Couzin et al.) ;
— self-organisation phenomena studied by Ilya Prigogine or Hermann Haken.

Thus, the boids model constitutes a bridge between computer simulation and the theory of complex systems.

Annexe B.7 Equivalent Continuous Model

In continuous form, the equation of motion of a boid is written as

$$\frac{dX_i}{dt} = V_i,$$

$$\frac{dV_i}{dt} = k_c(\bar{X}_i - X_i) + k_a(\bar{V}_i - V_i) - k_s \sum_{j \in \mathcal{N}_i} \frac{X_j - X_i}{\|X_j - X_i\|^2 + \varepsilon}.$$

This nonlinear, self-regulated system is simulated numerically by discrete integration.

Annexe B.8 Extension : Radial Danger Field

To incorporate escape behaviour in response to a localised danger, an external field F_{danger} centred at X_0 can be added.

One possible radial formulation is

$$F_{\text{danger},i} = k_d e^{-\frac{\|X_i - X_0\|^2}{2\sigma^2}} \frac{X_i - X_0}{\|X_i - X_0\| + \varepsilon}.$$

This field is intended to generate an outward escape response when the school reacts to a localised threat such as a predator.

The combination of internal forces and the danger field may produce collective avoidance, dispersion and subsequent realignment.

This extension remains a theoretical model ; it is not part of the calibrated experimental comparison presented below.

Annexe B.9 Qualitative Interpretation

The boids model illustrates the transition from the *local to the global* : group cohesion does not necessarily result from central command, but can emerge from distributed interactions.

It is a paradigmatic example of *emergent behaviour*.

Annexe C – Implementation and Visualisation of the Fish School

Annexe C.1 Simulation Algorithm

The simulation algorithm is based on behavioural modelling of the Aoki (1982) and Reynolds (1987) type.

Each fish interacts with nearby individuals according to three principles :

1. **Repulsion** : avoid collisions with fish that are too close ;
2. **Orientation** : align with the average direction of neighbouring individuals ;
3. **Attraction** : move closer to the local centre of the group to maintain cohesion.

The simulated positions can then be represented as a dynamic graph, allowing the comparison of simulated and experimentally observed spatial structures.

Annexe C.2 Simulation of Behavioural Forces

The Python implementation separates the simulation of individual movement from the construction and analysis of graphs.

The function `step_boids()` applies the local behavioural rules to update the simulated positions and velocities.

Three fundamental forces are considered : separation, alignment and cohesion.

The coefficients associated with these forces determine their relative importance in the resulting movement.

The implementation also provides an inertial reference model through the function `step_persistence()`.

This reference model continues the movement using the initial displacement, without introducing the three behavioural interactions.

Annexe C.3 Initialisation from Experimental Data

The simulations are initialised using experimentally observed positions from the fish-school dataset.

Initial velocities are estimated from consecutive observed positions.

This distinction is important because the fields `vx` and `vy` in the dataset represent directional information and should not automatically be interpreted as physical velocities expressed in distance per unit of time.

The simulations therefore use the observed displacements to initialise the predictive model.

Annexe C.4 Management of Spatial Boundaries

Spatial boundaries may be introduced in a general fish-school simulation to represent the walls of an aquarium. In such a model, when a fish reaches the limits

$$(x_{\min}, x_{\max}, y_{\min}, y_{\max}),$$

its velocity can be reversed or corrected to prevent it from leaving the simulation area.

This constitutes a possible boundary condition for the theoretical model.

The experimental comparisons presented below are instead interpreted through the numerical simulation implemented in the project, without attributing unverified physical dimensions to the observation area.

Annexe C.5 Animation of the Fish School

Dynamic visualisation is produced using the Python `matplotlib` library.

The graphical interface displays three representations of the same group :

1. the inertial reference simulation ;
2. the experimentally observed fish school ;
3. the interactive Boids simulation.

The comparison is performed on a common cohort of tracked fish, allowing their positions and collective organisation to be compared at the same prediction horizon.

Annexe C.6 Analysis of Interindividual Distances and Monitoring Graphs

At each selected prediction horizon, interindividual distances can be calculated from the positions of the fish.

The distance to the nearest neighbour is

$$d_i^{\mathrm{NN}}(t) = \min_{j \neq i} \|X_i(t) - X_j(t)\|.$$

The mean nearest-neighbour distance is

$$\overline{d}_{\mathrm{NN}}(t) = \frac{1}{N} \sum_{i=1}^{N} d_i^{\mathrm{NN}}(t).$$

These measurements provide indicators of the spatial organisation of the school.

They can be compared between the observed trajectories, the Boids simulation and the inertial reference model.

Annexe C.7 Visualisation of Collective Behaviour

The program generates a dynamic representation of the fish and their movement directions.

Fish positions are displayed graphically, and their local relationships may be represented using neighbourhood links.

The display evolves over time, allowing the observation of directional alignment, local dispersion and regrouping movements.

Annexe C.8 Interactive Interface and User Control

An interactive graphical interface was developed using Python, Matplotlib and `matplotlib.widgets`.

It provides a visual environment for exploring collective fish movement, adjusting simulation parameters and comparing predictions with experimental observations.

Organisation of the Interface

The interface contains three synchronised panels arranged horizontally :

— **Inertia (left) :** displays the inertial reference model, in which the initial movement is extrapolated without explicit behavioural interactions.

— **Real observations (centre) :** displays the actual fish positions and movement directions extracted from the experimental dataset.

— **Interactive Boids (right) :** displays the simulated trajectories obtained from the local rules of separation, alignment and cohesion.

All three panels use the same coordinate system and the same cohort of tracked fish. This arrangement allows the predicted and observed spatial configurations to be compared directly.

Fish are represented by triangular markers oriented according to their direction of movement.

The triangles therefore encode two complementary properties : the position of each fish and its instantaneous movement direction.

When spatial edges are enabled, lines connect fish belonging to the same spatial neighbourhood, providing a graphical representation of local network connectivity.

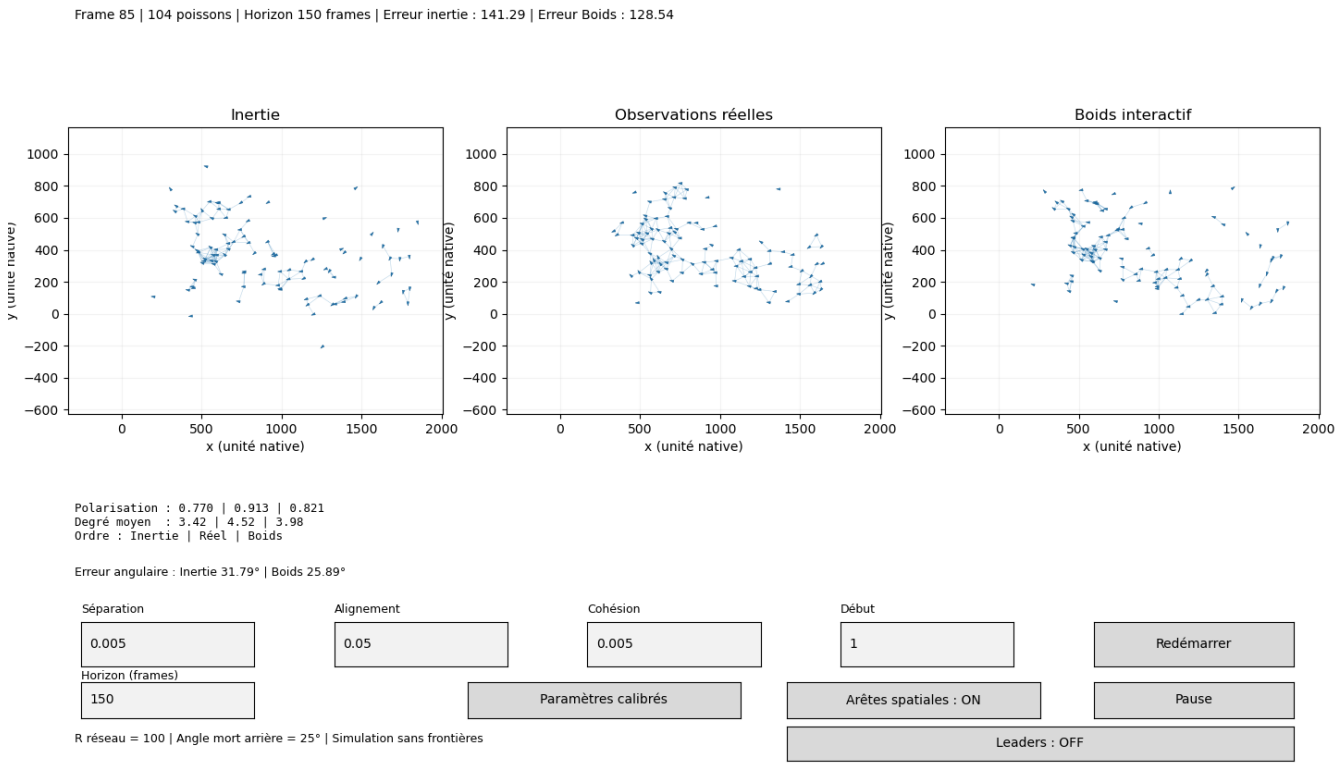


FIGURE 8 – Interactive Matplotlib interface for comparing the inertial reference model (left), real fish observations (centre) and Boids simulation (right). Oriented triangular markers represent fish positions and movement directions. Spatial edges illustrate local neighbourhood relationships. The lower part of the interface contains simulation parameters, evaluation metrics and interactive controls.

Simulation Information

The upper part of the interface displays the current frame, the number of fish included in the comparison and the selected prediction horizon.

The displayed number of fish corresponds to the cohort retained for the selected simulation interval, rather than necessarily to all fish detected in the original dataset.

The horizontal and vertical axes represent the spatial coordinates of the fish in native dataset units.

These coordinates are not interpreted as physical distances in metres or centimetres because their physical calibration has not been established.

Quantitative Indicators

The interface provides several indicators for comparing the observed and simulated collective behaviour.

— **Position error :** measures the average distance between predicted and observed fish positions. Lower values indicate closer geometric agreement.

— **Angular error :** measures the average difference between predicted and observed movement directions, in degrees. Lower values indicate more accurate directional predictions.

— **Collective polarisation :** measures the degree of alignment of movement directions within the fish school. Its value ranges from 0 to 1, where values close to 1 indicate strongly aligned movement.

— **Mean graph degree :** represents the average number of spatial neighbours per fish in the corresponding graph.
Higher values indicate a greater number of local connections, under the selected distance threshold.

The interface displays polarisation and mean degree for the inertial, observed and Boids configurations.

It also reports position and angular errors for the two predictive models relative to the real observations.

Illustrative Comparison

Figure 8 shows an example at frame 85, with 104 tracked fish and a prediction horizon of 150 frames.

For this displayed state, the position error is 141.29 for the inertial reference model and 128.54 for the Boids simulation.

The corresponding angular errors are 31.79 and 25.89 degrees, respectively.

The displayed polarisation values are 0.770 for inertia, 0.913 for the observations and 0.821 for Boids.

The mean graph degrees are 3.42, 4.52 and 3.98, respectively.

In this example, the Boids simulation provides a smaller positional and angular discrepancy than the inertial reference model.

However, these values correspond to a single displayed state. They should not be interpreted as evidence that Boids systematically outperforms inertia over all prediction horizons or observation windows.

The experimental validation presented in Appendix D provides a more systematic comparison.

Annexe C.9 User Manual for the Visual Interface

In the two-dimensional simulation model developed in Python, several parameters control the behaviour of the school and the visualisation of its evolution.

These parameters modulate local interactions, simulation duration and the display of graph-related information.

Behavioural Parameters

The interface provides editable fields for the three main behavioural coefficients :

- **Separation coefficient :** controls the intensity of the short-range repulsive force, which tends to prevent fish from approaching too closely.
- **Alignment coefficient :** governs the tendency of fish to adjust their movement directions towards those of neighbouring fish.
- **Cohesion coefficient :** controls the attraction of fish towards the local centre of their neighbourhood.

The theoretical model also defines interaction radii for separation, alignment and cohesion.

These radii determine which fish are considered neighbours for each behavioural interaction.

They should be distinguished from the editable coefficients displayed in the interface, which control the strength of the corresponding forces.

Simulation Controls

The graphical interface includes the following interactive controls :

- **Start frame :** selects the initial frame from which the simulation comparison is performed.
- **Prediction horizon :** determines the number of frames considered in the simulation.
- **Calibrated parameters :** applies a predefined configuration of behavioural coefficients obtained from the calibration procedure.
- **Restart :** reinitialises the simulation using the selected settings.
- **Pause :** temporarily stops the animation, allowing the current spatial configuration to be examined.
- **Spatial edges :** enables or disables the display of local graph connections.
- **Leaders :** enables or disables the display of potential leaders identified by the temporal analysis, when corresponding information is available.

The spatial graph is constructed using a neighbourhood distance threshold of $R = 100$ in the displayed configuration.

The interface also indicates a rear blind angle of 25°, associated with the visibility modelling assumptions of the project.

The simulation shown in the figure operates without explicit spatial boundaries.

Execution of the Interface

The graphical interface is implemented in the Python module

```
src/boids_interface.py
```

From the root directory of the project repository, it can be launched using :

```
python -m src.boids_interface
```

The required Python dependencies are listed in the repository's `requirements.txt` file.

The original fish-school dataset must also be available at the location expected by the program.

Annexe C.10 Graphical Analysis and Export

The project produces numerical results and graphical representations for several prediction horizons.

The generated figures include comparisons of spatial neighbourhoods, collective polarisation, angular errors, mean degree and nearest-neighbour distances.

These visualisations provide a complementary interpretation of the numerical indicators.

The interactive interface additionally allows the user to inspect the evolution of the three models at corresponding simulation frames.

It therefore connects the numerical evaluation of predictive performance with the visual interpretation of collective movement and graph structure.

Annexe C.11 Difficulties Encountered

The experimental data introduce several limitations for simulation and validation.

First, the number of detected fish varies between frames.

Second, tracking identifiers do not guarantee permanent biological identity throughout the complete recording.

Third, the spatial coordinates are expressed in native dataset units whose physical calibration is not established.

Consequently, the simulation is evaluated on cohorts of fish continuously observed within the selected windows.

The results describe the performance of a computational model under these conditions, rather than a complete biological reconstruction of fish behaviour.

Annexe D – Experimental Calibration and Validation of the Boids Model

Annexe D.1 Experimental Objective

The purpose of the experimental analysis is to evaluate whether the Boids model can reproduce the observed movements and spatial organisation of a fish school.

The simulations are compared with the experimental trajectories using a common initial state.

The main question is whether local rules of separation, alignment and cohesion provide a measurable improvement over a simple inertial reference model.

Annexe D.2 Experimental Dataset and Windows

The experiments use the *Fish Schooling Data Subset* from Oregon State University.

The dataset contains 5,000 frames recorded at 30 frames per second.

For the predictive comparison, three temporal windows are considered :

— Frames 1–32 : calibration and initial evaluation ;

— Frames 2000–2031 : independent validation ;

— Frames 4000–4031 : independent validation.

The simulations are initialised from the first two observed positions.

Predictions are then evaluated at horizons of 5, 10, 20 and 30 frames.

Only fish with continuous observations throughout the corresponding window are retained.

The resulting cohort sizes are 184 fish for the training window, 190 fish for the first validation window and 232 fish for the second validation window.

Annexe D.3 Inertial Reference Model

The inertial model provides a baseline against which the behavioural simulation can be evaluated.

For each fish, the initial displacement is estimated from two consecutive observations :

$$\widehat{V}_i(1) = X_i(2) - X_i(1).$$

The subsequent position is predicted by maintaining this displacement :

$$\widehat{X}_i(t+1) = \widehat{X}_i(t) + \widehat{V}_i(1).$$

This model does not include separation, alignment or cohesion.

Its purpose is to determine whether the introduction of local interactions improves the prediction.

Annexe D.4 Calibration of Behavioural Parameters

The Boids model is controlled by three principal coefficients :

$$\theta = (k_s, k_a, k_c),$$

where k_s, k_a and k_c correspond respectively to separation, alignment and cohesion.

A first calibration selected

$$\boxed{k_s = 0, \qquad k_a = 0.15, \qquad k_c = 0.}$$

This configuration corresponds to an alignment-only model.

A second optimisation explored 392 combinations of the three behavioural coefficients.

The calibration was performed on the training window only, using the same spatial neighbourhood radius

$$R = 100.$$

Two optimisation objectives were considered :

1. maximising the mean spatial edge Jaccard similarity ;
2. minimising the position error at a prediction horizon of 30 frames.

Annexe D.5 Optimised Parameter Configurations

The optimisation produced different parameter configurations depending on the evaluation criterion.

Configuration	k_s	k_a	k_c
Initial calibration	0	0.15	0
Best spatial Jaccard	0.005	0.05	0.005
Best position error	0.025	0.30	0.01

TABLE 4 – Behavioural coefficients selected by the calibration procedures.

The results show that the preferred combination depends on the objective.

The best configuration for reproducing the graph structure is not necessarily the best configuration for minimising the geometric prediction error.

Annexe D.6 Evaluation Metrics

Annexe D.6.1 Spatial Edge Jaccard Similarity

The spatial graph is constructed from the fish positions using a distance threshold $R = 100$.

For a given prediction horizon, let E_{real} denote the set of observed edges and E_{sim} the set of simulated edges.

The Jaccard similarity is

$$J(E_{\text{real}}, E_{\text{sim}}) = \frac{|E_{\text{real}} \cap E_{\text{sim}}|}{|E_{\text{real}} \cup E_{\text{sim}}|}.$$

A value close to 1 indicates that the simulated and observed graphs contain similar neighbourhood relationships.

A value close to 0 indicates limited overlap.

This measure evaluates structural similarity rather than exact geometric correspondence.

Annexe D.6.2 Mean Position Error

The mean position error is defined as

$$E_{\text{position}}(t) = \frac{1}{N} \sum_{i=1}^{N} \left\| X_i^{\text{sim}}(t) - X_i^{\text{real}}(t) \right\|.$$

This metric measures the average geometric difference between simulated and observed positions.

The error is expressed in native dataset units.

A smaller value indicates a more accurate geometric prediction.

Annexe D.6.3 Collective Polarisation

Collective polarisation measures the degree of directional alignment within the school.

For the unit movement direction $\widehat{V}_i(t)$ of each moving fish,

$$P(t) = \left\| \frac{1}{N} \sum_{i=1}^{N} \widehat{V}_i(t) \right\|.$$

A value close to 1 corresponds to strongly aligned movement.

A value close to 0 corresponds to weak overall directional coherence.

The difference between simulated and observed polarisation provides an additional measure of collective behaviour.

Annexe D.6.4 Mean Angular Error

The angular difference between the simulated and observed movement directions is

$$\theta_i(t) = \arccos\left(\frac{V_i^{\text{real}}(t) \cdot V_i^{\text{sim}}(t)}{\|V_i^{\text{real}}(t)\| \|V_i^{\text{sim}}(t)\|} \right).$$

The mean angular error is

$$E_{\text{angular}}(t) = \frac{1}{N_{\text{valid}}} \sum_{i=1}^{N_{\text{valid}}} \theta_i(t).$$

Only pairs with nonzero movement vectors are considered.

The resulting angle is expressed in degrees.

Annexe D.6.5 Structural Graph Metrics

Additional indicators are calculated from the spatial graphs.

The mean degree is

$$\overline{k}(t) = \frac{1}{N} \sum_{i=1}^{N} k_i(t),$$

where $k_i(t)$ denotes the number of neighbours of fish i.

The nearest-neighbour distance is also measured for each fish.

These metrics make it possible to compare the simulated local structure with the experimentally observed structure.

Annexe D.7 Validation on Independent Windows

The selected parameter configurations were evaluated on two temporal windows not used during calibration.

At a prediction horizon of 30 frames, the following results were obtained.

Window	Model	Jaccard	Position error
2000	Inertia	0.5046	40.04
2000	Initial calibration	0.5290	39.30
2000	Jaccard-optimised	0.5182	39.28
2000	Position-optimised	0.5321	38.09
4000	Inertia	0.5198	37.36
4000	Initial calibration	0.5407	35.94
4000	Jaccard-optimised	0.5458	35.54
4000	Position-optimised	0.5384	35.91

TABLE 5 – Comparison of the inertial reference and the calibrated Boids configurations at a prediction horizon of 30 frames. Position errors are expressed in native dataset units.

On the first validation window, the position-optimised model achieves a spatial Jaccard similarity of 0.5321, compared with 0.5046 for the inertial reference.

Its mean position error is 38.09, compared with 40.04 for inertia.

On the second validation window, the Jaccard-optimised model achieves a spatial similarity of 0.5458, compared with 0.5198 for inertia.

Its mean position error is 35.54, compared with 37.36 for the inertial model.

Annexe D.8 Interpretation of the Results

The experimental results show that local behavioural interactions can improve the reproduction of the observed fish-school structure over the selected prediction horizons.

However, the improvement remains relatively modest.

The results also reveal that the calibration criterion matters : a model optimised for spatial connectivity does not necessarily minimise the position error on every validation window.

Furthermore, a configuration selected during training does not systematically outperform all alternatives on unseen data.

These observations illustrate the importance of evaluating the simulation with several complementary metrics.

Annexe D.9 Experimental Limitations

Several limitations must be considered when interpreting the results.

First, the validation is based on only two independent temporal windows.

Second, the fish included in each comparison are selected according to the continuity of their tracking identifiers over the evaluation period.

Third, the physical scale of the position coordinates is not established.

Finally, the similarity between observed and simulated trajectories does not demonstrate that the real fish follow the Boids rules.

The results therefore support a limited predictive comparison, rather than a causal explanation of biological collective behaviour.

Annexe E – Temporal Leadership Detection and Collective Behaviour

Annexe E.1 Leadership Hypotheses

Several hypotheses may explain the observed movements of a fish school : erratic or genuinely random movements, or movements dependent on certain individuals whose directional changes precede those of nearby fish.

These individuals might be located at the front of the school, at its centre or on its periphery, where they may respond to environmental changes.

Different collective trajectories may therefore be obtained from different hypotheses.
The relevance of each hypothesis must ultimately be tested by comparison with observed facts.

Annexe E.2 Objective of the Temporal Analysis

The purpose of the leadership analysis is to identify individuals whose directional changes systematically precede similar changes among neighbouring fish.
The analysis is based on the temporal evolution of the observed trajectories.
It does not assume that the school possesses a permanent leader or a centralised decision-making structure.
Instead, leadership is treated as a potential and measurable temporal relationship.

Annexe E.3 Construction of Turning Signals

For each fish i, the observed displacement between two consecutive frames is defined as

$$V_i(t) = X_i(t+1) - X_i(t).$$

The movement direction is calculated using

$$D_i(t) = \operatorname{atan2}\left(V_i^y(t), V_i^x(t)\right).$$

The angular variation between consecutive movement directions is

$$\Delta D_i(t) = \operatorname{wrap}\left(D_i(t+1) - D_i(t)\right),$$

where the wrapping operation returns the signed angular difference in the interval

$$[-\pi, \pi].$$

This transformation makes it possible to study changes in direction rather than absolute positions alone.

Annexe E.4 Temporal Correlation Between Fish

To evaluate whether the directional changes of fish i precede those of fish j, the turning signals are compared with a temporal lag τ.
The corresponding correlation is

$$\rho_{ij}(\tau) = \operatorname{corr}\left(\Delta D_i(t), \Delta D_j(t+\tau)\right).$$

Positive values of τ represent a situation in which the turning signal of fish i is compared with a later turning signal of fish j.
The analysis considers lags between 1 and 10 frames :

$$\tau \in \{1, 2, \ldots, 10\}.$$

At a recording frequency of 30 frames per second, this corresponds to delays between approximately 0.033 and 0.333 seconds.
The strongest lagged correlation is defined as

$$\rho_{ij}^{\max} = \max_{\tau \in \{1,\ldots,10\}} \rho_{ij}(\tau).$$

Annexe E.5 Identification of Potential Followers

Two fish are considered spatial neighbours when their mean proximity over a phase satisfies the selected criterion.
The spatial distance threshold is

$$R = 100.$$

In the implementation, a pair is retained when the fish remain within this distance for at least half of the observations in the corresponding phase.
A potential directional relationship from fish i to fish j is retained when the following conditions are satisfied :

$$\rho_{ij}^{\max} \geq 0.6,$$

and

$$\rho_{ij}^{\max} - \rho_{ij}(0) \geq 0.1.$$

The second condition requires the lagged correlation to exceed the simultaneous correlation by at least 0.1.
The implementation also requires a minimum of 12 valid observation pairs for the correlation calculation.
These conditions reduce the number of weak or poorly supported temporal associations.
They do not establish causal influence.

Annexe E.6 Influence Score

For each fish i, an influence score is calculated from the accepted lagged correlations :

$$I_i = \sum_{j \in \mathcal{F}_i} \rho_{ij}^{\max},$$

where $\mathcal{F}_i$ denotes the set of potential followers identified for fish i.
The number of followers is

$$n_i^{\text{followers}} = |\mathcal{F}_i|.$$

An individual is classified as a potential leader during a phase when

$$n_i^{\text{followers}} \geq 2.$$

This classification is based on a numerical criterion and should not be interpreted as proof of biological leadership.

Annexe E.7 Temporal Phases and Persistence

The analysis is performed on 150 consecutive frames starting at frame 2000.
The observation period is divided into five phases of 30 frames each.
For each fish, a binary indicator $L_i^{(p)}$ is defined :

$$L_i^{(p)} = \begin{cases} 1, & \text{if fish } i \text{ is a potential leader in phase } p, \\ 0, & \text{otherwise.} \end{cases}$$

The persistence score is

$$P_i = \frac{1}{5} \sum_{p=1}^{5} L_i^{(p)}.$$

A fish is considered a persistent potential leader when it is classified as a leader in at least three of the five phases :

$$\sum_{p=1}^{5} L_i^{(p)} \geq 3.$$

This definition distinguishes occasional temporal precedence from repeated identification across several phases.

Annexe E.8 Experimental Results

The temporal analysis identified several individuals whose directional changes preceded those of neighbouring fish during particular phases.
The tracking identifier 105760 was classified as a potential leader in three of the five phases.
Its persistence score is

$$P_{105760} = \frac{3}{5} = 0.60.$$

The corresponding cumulative influence score is approximately

$$I_{105760} = 5.7558.$$

This individual is therefore identified as a persistent potential leader under the selected numerical criteria.
Other fish were identified as potential leaders in one or two phases, indicating that temporal precedence may vary during the observation period.

Fish ID	Leader phases	Persistence	Total influence
105760	3	0.60	5.7558
104628	2	0.40	4.0692
105469	2	0.40	3.3261
105424	2	0.40	2.5783
105561	2	0.40	2.7288

TABLE 6 – Examples of potential leaders identified by the temporal correlation analysis. The identifiers correspond to tracking labels within the selected observation window.

Annexe E.9 Relationship with Graph Centrality

The leadership analysis complements the graph-theoretical study presented in the main report.

Graph centrality measures describe the structural position of a node within a network.

For example, betweenness centrality identifies nodes located on a large proportion of shortest paths.

However, structural centrality and temporal leadership are not equivalent concepts.

A fish may occupy a central position in a proximity graph without systematically preceding the directional changes of its neighbours.

Conversely, a fish may exhibit temporal precedence without being the most central node in the network.

The two approaches therefore provide complementary descriptions of collective organisation.

Annexe E.10 Comparison with the Boids Model

The Boids model is based on decentralised local rules.

In its standard formulation, no permanent leader is required to coordinate the group.

Nevertheless, some individuals may temporarily precede others in their directional changes.

The temporal leadership analysis therefore provides an additional way to study the organisation of collective movement.

It may help distinguish the existence of temporary directional precedence from the stronger hypothesis of permanent hierarchical leadership.

The presence of a temporally leading individual does not invalidate the possibility of decentralised collective behaviour.

Annexe E.11 Limitations of Leadership Detection

The interpretation of the leadership results requires several precautions.

First, a positive lagged correlation does not establish a causal relationship.

Second, two fish may react to the same external stimulus without one influencing the other.

Third, the tracking identifiers do not guarantee permanent biological identity throughout the entire dataset.

Fourth, the results depend on the selected distance threshold, correlation threshold, temporal lags and phase duration.

Finally, the analysis is performed on a restricted observation window.

The detected leaders must therefore be interpreted as *potential temporal leaders*, rather than confirmed biological leaders.

Annexe E.12 Interpretation of Collective Behaviour

The results illustrate the complexity of collective organisation within a fish school.

The movements of the group may result from local interactions, temporary directional precedence and changes in spatial organisation.

The graph-theoretical analysis describes the structure of the school, while the temporal correlation analysis examines the ordering of directional changes.

Together, these approaches provide complementary information about collective dynamics.

They do not, however, establish a unique causal mechanism responsible for the observed movements.

Annexe F – General Discussion of the Simulation Extensions

Annexe F.1 Relationship Between the Graph Models and the Simulations

The graph analysis and the Boids simulations address different but complementary aspects of the problem.

The spatial graph describes the proximity relationships between fish at a given instant.

The directed visibility graph introduces an additional hypothesis concerning the potential perception of neighbouring individuals.

The Boids model attempts to reproduce the evolution of positions using local behavioural rules.

Finally, the leadership analysis investigates whether directional changes of certain individuals temporally precede those of their neighbours.

These approaches should not be confused.

A proximity edge does not directly represent a biological interaction.

A visibility arc does not demonstrate that a fish is actually attending to another individual.

Likewise, a simulated trajectory that resembles an observed trajectory does not establish that the simulated forces correspond to the real biological mechanisms.

Annexe F.2 Principal Findings

The experimental comparisons show that the calibrated Boids models can improve the prediction of spatial relationships and positions relative to the inertial baseline on the selected validation windows.

However, the magnitude of the improvement is limited.

The optimisation also shows that the choice of evaluation criterion influences the selected behavioural parameters.

The temporal leadership analysis identifies individuals exhibiting repeated directional precedence, including one tracking identifier classified as a persistent potential leader in three of five phases.

These findings provide descriptive evidence of heterogeneous collective dynamics.

They do not demonstrate the existence of a permanent hierarchy or a specific biological mechanism.

Annexe F.3 Perspectives

Several extensions could improve the analysis.

A first possibility would be to evaluate the simulations on a larger number of independent temporal windows.

A second would be to investigate the sensitivity of the leadership results to the selected thresholds and lags.

A third would be to compare additional behavioural models with the same experimental trajectories.

Finally, a physical calibration of the spatial coordinates would improve the interpretability of the distances and interaction radii.

These developments would provide a more robust assessment of the relationship between local behavioural rules and observed collective movement.

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Appendix H – Software Architecture and Python Modules

H.1. General Architecture

The project is implemented in Python and organised into independent modules covering data processing, graph construction, network analysis, numerical simulation, visualisation and experimental validation.

The source code is publicly available at :

`github.com/Charles-Scannavino/Projet_Graph_Analytics`

The repository separates the scientific analysis of experimental fish-schooling networks from the complementary Boids simulation and leadership analysis.

The main libraries include NumPy, SciPy, NetworkX, Matplotlib and ijson. The pytest framework is used for automated testing.

H.2. Data Loading and Graph Construction

`data_loader.py` Loads experimental observations from the JSON dataset. It extracts fish identifiers, positions and velocity components, and checks the structure of the input data.

`graph_builder.py` Constructs an undirected spatial proximity graph. Each observed fish is represented by a vertex. Two vertices are connected when their Euclidean distance is smaller than the selected neighbourhood radius. Distances are stored as edge attributes.

`visibility_graph.py` Constructs a directed visibility graph using spatial proximity and an angular visibility constraint. The reference configuration uses a total rear blind sector of 25 degrees, with swimming direction approximated from the recorded velocity vector.

H.3. Graph Analytics and Community Detection

`graph_metrics.py` Computes network characteristics, including graph size, density, connected components, topological distances and centrality measures.

`community_detection.py` Applies Louvain community detection to identify groups of densely interconnected fish. It produces community memberships and modularity measurements.

`temporal_analysis.py` Constructs and analyses networks at selected experimental frames. It measures changes in network structure over time and exports the corresponding numerical results.

`sensitivity_analysis.py` Studies the influence of the neighbourhood radius on graph density, connectivity, clustering, community structure and modularity.

H.4. Boids Simulation and Calibration

The Boids extension investigates whether local separation, alignment and cohesion rules can reproduce observed collective motion.

`trajectory_audit.py` Examines the continuity of tracked fish identifiers and checks trajectory data quality across selected experimental windows.

`boids_pilot.py` Implements the experimental Boids simulation and compares predicted movements with a simple inertial reference model.

`boids_behavior_metrics.py` Provides behavioural indicators used to characterise collective movement and compare observed and simulated dynamics.

`boids_structural_metrics.py` Provides graph-based structural indicators for evaluating the spatial organisation of simulated fish schools.

`compare_boids_graphs.py` Compares experimental and simulated proximity and visibility networks using structural metrics and edge-set similarity.

`calibrate_boids.py` Explores combinations of behavioural coefficients on the calibration window and selects a parameter configuration according to the defined objective.

`optimize_boids_weights.py` Provides an additional optimisation workflow for the behavioural coefficients of the simulation model.

`validate_boids_windows.py` Evaluates simulation performance across several experimental windows and prediction horizons.

`validate_calibrated_boids.py` Evaluates calibrated parameters on independent validation windows without repeating calibration.

`validate_optimized_weights.py` Provides a validation workflow for the optimised behavioural weights.

`summarize_boids_validation.py` Consolidates experimental validation results into summary tables for comparison and reporting.

H.5. Leadership Analysis

`detect_fish_leaders.py` Implements the identification of potential leaders from temporal fish movement data.

`compare_fish_leadership.py` Compares leadership indicators and their evolution across experimental configurations.

`leader_display.py` Provides visualisation utilities for displaying potential leaders within the fish-school interface.

Leadership indicators are interpreted as exploratory computational measures. They do not establish biological leadership or causal influence.

H.6. Visualisation and Interactive Interface

`visualize.py` Generates the reference spatial and visibility network figures, including Louvain communities. Fish are displayed using oriented markers based on their recorded movement directions. Figures are exported in PNG and SVG formats.

`plot_temporal.py` Produces figures illustrating the temporal evolution of graph characteristics.

`plot_boids_comparison.py` Generates comparative figures for experimental, inertial and Boids simulation results.

`boids_interface.py` Implements the interactive graphical interface used to inspect fish-school simulations. It provides controls for behavioural parameters, simulation playback and graph-related displays.

H.7. Automated Testing

The repository contains a dedicated `tests/` directory with automated tests.
The following test modules are included :

- `test_graph_models.py`
- `test_graph_metrics.py`
- `test_community_detection.py`
- `test_temporal_analysis.py`
- `test_sensitivity_analysis.py`
- `test_visualize.py`
- `test_plot_temporal.py`
- `test_boids_pilot.py`
- `test_boids_structural_metrics.py`
- `test_compare_boids_graphs.py`
- `test_compare_fish_leadership.py`
- `test_leader_display.py`

These tests cover graph construction, network metrics, community detection, temporal analysis, simulation components and selected visualisation functions.

The package also includes `src/__init__.py`, which identifies the source directory as a Python package.

H.8. Execution and Reproducibility

The project requires a compatible Python environment, the dependencies listed in `requirements.txt`, and access to the original experimental dataset.

The dataset is expected at :

```
data/schooling_frames.json
```

The principal analyses can be reproduced from the repository root using the following commands :

```
python -m src.graph_metrics
python -m src.community_detection
python -m src.visualize
python -m src.temporal_analysis
python -m src.plot_temporal
python -m src.sensitivity_analysis
```

The interactive simulation can be launched using :

```
python -m src.boids_interface
```

Automated tests can be executed using :

```
python -m pytest -q
```

Generated outputs are stored under `results/`, including numerical results, experimental summaries and graphical figures.

The complete source code, execution instructions, dependencies and available results are maintained in the GitHub repository.

H.9. Scope and Scientific Reproducibility

The modular implementation separates the construction of experimental networks from their analysis and from predictive simulation.

This separation makes it possible to examine the effects of graph definitions, temporal sampling and neighbourhood parameters without confusing them with the assumptions of the Boids model.

The original experimental observations remain the reference data. Simulated trajectories and derived network measures are interpreted as computational approximations rather than direct biological observations.

The repository provides the code required to inspect the methodology and reproduce the analyses, subject to the availability of the original dataset and a compatible environment.

Appendix I – Use of Artificial Intelligence

Artificial Intelligence tools, primarily ChatGPT, were used as supporting resources throughout the project, particularly for Python programming assistance, debugging, code documentation, and improving the clarity of technical explanations.

AI assistance also helped explore possible analytical approaches and organize the presentation of the results. However, the research question, methodological choices, graph construction parameters, and interpretation of the findings remained under the responsibility of the students.

The generated suggestions were reviewed, adapted, and checked against the dataset, Python implementations, and experimental results. AI was therefore used as a complementary development and learning tool, rather than as a substitute for independent scientific reasoning and critical analysis.